Roadmap. Page order ≠ teaching order. §36 (pp. 1–10): Lorentz-invariant spin-$\frac{1}{2}$ Lagrangians from Weyl fields (Majorana, Dirac; β, C, γ_5, $S^{\mu\nu}$); solutions in LecNote18. §47 (pp. 10–12): traces, taught in Class 25 for LecNote20 p. 6. §43 (pp. 13–16): $Z_0(\bar\eta,\eta)$. §44 (pp. 16–21, skipped): Grassmann calculus behind p. 15. §45 (pp. 21–22): Yukawa Z, continued in LecNote18 pp. 1–2. Supplements: SpinorRepresentations, LorentzGroup, FeynmanPropagator, DiracSpinors, GammaTraces, FermionSigns, DiracFeynmanRules.

Lec 17

Dirac Equation & Spinor Fields

With all the invariant symbols we can construct a Lorentz invariant Lagrangian density for spinor fields.

as $(i\psi^\dagger\bar\sigma^\mu\partial_\mu\psi)^\dagger = i\partial_\mu\psi^\dagger\bar\sigma^\mu\partial_\mu\psi + -i\partial_\mu(\psi^\dagger\bar\sigma^\mu\psi)$

↑ surface terms.

and $(m\psi\psi)^\dagger = m^*\psi^\dagger\psi^\dagger$, we have

$$\mathcal{L} = i\psi^\dagger\bar\sigma^\mu\partial_\mu\psi - \frac{1}{2}|m|(\psi\psi + \psi^\dagger\psi^\dagger)$$

↑ complex phase absorbed in $\psi \to \psi e^{\frac{i\alpha}{2}}$

E.o.M.s is then

$$0 = -\frac{\delta S}{\delta\psi^\dagger} = -i\bar\sigma^\mu\partial_\mu\psi + m\psi^\dagger$$

$$0 = -\frac{\delta S}{\delta\psi} = -\left(\frac{\delta S}{\delta\psi^\dagger}\right)^\dagger$$

$$= +i(\bar\sigma^\mu)^{\dot a a *}\partial_\mu\psi^\dagger_{\dot c} + m\psi^a$$

$$= +i(\bar\sigma^{\mu\dot c a})\partial_\mu\psi^\dagger_{\dot c} + m\psi^a$$

$$\Rightarrow -0 = -i\,\sigma^\mu_{a\dot c}\partial_\mu\psi^{\dot c\dagger} + m\psi_a$$

↑ $\epsilon_{ba}\to\epsilon_{ab}$ ↑ $\dot c_{\dot c}\to\dot c^{\dot c}$

$= i\bar\sigma^\mu\partial_\mu\psi^\dagger + m\psi$

We see the EOM of ψ and $\psi^\dagger$ are mixed.

$$\begin{pmatrix} m\delta_a{}^c & -i\sigma^\mu_{a\dot{c}}\partial_\mu \\ -i\bar{\sigma}^{\mu\dot{a}c}\partial_\mu & m\delta^{\dot{a}}_{\dot{c}} \end{pmatrix}\begin{pmatrix} \psi_c \\ \psi^{\dagger\dot{c}} \end{pmatrix} = 0$$

Traditionally, we define Gamma Matrices (How people use to do calculations)

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu_{a\dot{c}} \\ \bar{\sigma}^{\mu\dot{a}c} & 0 \end{pmatrix}$$

It is ~~hermitian~~ not hermitian, but Lorentz invariant ~~not~~

as $\bar{\sigma}$ and σ are 2. I. symbols that we constructed earlier.

Not that $\sigma^\mu = (I, \vec{\sigma})$, $\bar{\sigma}^\mu = (I, -\vec{\sigma})$

$$\{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}.$$

anticommutator as we already know from last before.

$$(\bar{\sigma}^\mu\sigma^\nu + \sigma^\nu\bar{\sigma}^\mu)^{\dot{a}}_{\dot{c}} = -2g^{\mu\nu}\delta^{\dot{a}}_{\dot{c}} \quad (1)$$

$$(\sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu)_a{}^c = -2g^{\mu\nu}\delta_a{}^c$$

We can then introduce a four component field

1 Should read $(\bar{\sigma}^\mu\sigma^\nu + \bar{\sigma}^\nu\sigma^\mu)^{\dot{a}}{}_{\dot{c}} = -2g^{\mu\nu}\delta^{\dot{a}}{}_{\dot{c}}$: both terms $\bar{\sigma}\sigma$, to carry ${}^{\dot{a}}{}_{\dot{c}}$. As written it fails ($\mu\nu = 01$ gives $2\sigma_1$). It is the lower-right block of $\{\gamma^\mu, \gamma^\nu\}$.

$\Psi = \begin{pmatrix} \psi_c \\ \psi^{\dagger c} \end{pmatrix}$ that is a majorana field.

Obeying the famous Dirac Equation. Which we haven't identified in Nature yet. And it doesn't conserve charge.

$$(-i\gamma^\mu \partial_\mu + m)\Psi = 0$$

Now, more typically, we talk about a Dirac Field.

Suppose that I have two field related by SO(2) transformation. (a, b) invariant under a^2+b^2

so $\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \Rightarrow \underbrace{\begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}}_{S} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$

keeps a^2+b^2 invariant as. $S^T S = \mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

such SO(2) transformation can also rewritten as U(1). (a phase).

$$\chi = \frac{1}{\sqrt{2}}(\psi_1 + i\psi_2)$$

$$\xi = \frac{1}{\sqrt{2}}(\psi_1 - i\psi_2).$$

Now the two-~~fold~~ left-handed spinor field Lagrangian density

$$\mathcal{L} = i\psi_1^\dagger \bar{\sigma}^\mu \partial_\mu \psi_1 + i\psi_2^\dagger \bar{\sigma}^\mu \partial_\mu \psi_2 - \frac{1}{2} m (\psi_1 \psi_1 + \psi_1^\dagger \psi_1^\dagger + \psi_2 \psi_2 + \psi_2^\dagger \psi_2^\dagger).$$

can be rewritten in such a new basis as

$$\mathcal{L} = i\chi^\dagger \bar{\sigma}^\mu \partial_\mu \chi + i\xi^\dagger \bar{\sigma}^\mu \partial_\mu \xi - m\chi\xi - m\xi^\dagger \chi^\dagger$$

And is invariant under

$$\chi \to e^{-i\alpha}\chi$$

$$\xi \to e^{+i\alpha}\xi$$

the equation of motion for this system is then

$$\begin{pmatrix} m\delta_a^c & -i\sigma^\mu_{ac}\partial_\mu \\ -i\bar{\sigma}^{\mu\dot{a}c}\partial_\mu & m\delta^{\dot{a}}_{\dot{c}} \end{pmatrix} \begin{pmatrix} \chi_c \\ \xi^{\dagger\dot{c}} \end{pmatrix} = 0$$

We can define the Dirac field

$$\Psi = \begin{pmatrix} \chi_c \\ \xi^{\dagger\dot{c}} \end{pmatrix}, \quad \bar{\Psi}^\dagger = (\chi_{\dot{c}}^\dagger, \xi^c)$$

Introducing a new matrix $\beta \equiv \begin{pmatrix} 0 & \delta^{\dot{a}}_{\dot{c}} \\ \delta_a^c & 0 \end{pmatrix}$

(this numerically equals γ^0 but meaning is different).

then

$$\bar{\Psi} \equiv \Psi^\dagger \beta = (\xi^a, \chi^\dagger_{\dot{a}})$$

that gives

$$\bar{\Psi}\Psi = \xi^a \chi_a + \chi^\dagger_{\dot{a}} \xi^{\dagger \dot{a}}$$, maintaining our convension

$$\bar{\Psi} \gamma^\mu \partial_\mu \Psi = (\xi^a, \chi^\dagger_{\dot{a}}) \begin{pmatrix} 0 & \sigma^\mu_{a\dot{c}} \partial_\mu \\ \bar{\sigma}^{\mu \dot{a} c} \partial_\mu & 0 \end{pmatrix} \begin{pmatrix} \chi_c \\ \xi^{\dagger \dot{c}} \end{pmatrix}$$

$$= \xi^a \sigma^\mu_{a\dot{c}} \partial_\mu \xi^{\dagger \dot{c}} + \chi^\dagger_{\dot{a}} \bar{\sigma}^{\mu \dot{a} c} \partial_\mu \chi_c$$

again IBP: $A \partial B = -(\partial A) B + \partial(AB)$.

Skip due to time

$$\rightarrow -(\partial_\mu \xi^a) \sigma^\mu_{a\dot{c}} \xi^{\dagger \dot{c}} + \partial_\mu (\quad)$$ surface term

$$= \xi^{\dagger \dot{c}} \sigma^\mu_{a\dot{c}} \partial_\mu \xi^a = \xi^\dagger_{\dot{c}} \bar{\sigma}^{\mu \dot{c} a} \partial_\mu \xi_a$$

number "—"

Recall (35.19)

$$\bar{\sigma}^{\mu \dot{a} a} = \varepsilon^{ab} \varepsilon^{\dot{a}\dot{b}} \sigma^\mu_{b\dot{b}}$$

Hence

$$\bar{\Psi}\gamma^\mu\partial_\mu\Psi = \chi^\dagger\bar{\sigma}^\mu\gamma_\mu\chi + \xi^\dagger\bar{\sigma}^\mu\gamma_\mu\xi + \partial_\mu(\text{surface}).$$

We then transform the original Lagrangian to a very compact form

$$\mathcal{L} = i\bar{\Psi}\gamma^\mu\partial_\mu\Psi - m\bar{\Psi}\Psi.$$

And such Lagrangian is invariant under

$$\Psi \to e^{-i\alpha}\Psi, \quad \bar{\Psi} \to e^{+i\alpha}\bar{\Psi}$$

which is the same as $\chi \to e^{-i\alpha}\chi$, $\xi^\dagger \to e^{-i\alpha}\xi^\dagger$

One can relate this to the charge, and the Noether current (of continuous symmetry)

$$j^\mu = \bar{\Psi}\gamma^\mu\Psi = \chi^\dagger\bar{\sigma}\chi - \xi^\dagger\bar{\sigma}\xi.$$

Here we could have an operation called Charge Conjugation.

$$C^{-1}\chi_a(x)C = \xi_a(x)$$

$$C^{-1}\xi_a(x)C = \chi_a(x)$$

And $C^{-1}\mathcal{L}(x)C = \mathcal{L}(x)$, the Lagrangian conserves charge.[1]

1 Should read: $\mathcal{L}$ is C-invariant (charge conjugation is a symmetry). C sends $j^\mu \to -j^\mu$; charge conservation is the $U(1)$ above. The two are independent: weak interactions violate C but conserve charge.

We can introduce the charge conjugation matrix

$$C = \begin{pmatrix} \varepsilon_{ac} & \\ & \varepsilon^{\dot{a}\dot{c}} \end{pmatrix}$$

To get $$C^{-1}\Psi C = \Psi^c = \begin{pmatrix} \xi_a \\ \chi^{+\dot{a}} \end{pmatrix}$$

notice that $$\bar{\Psi}^T = \begin{pmatrix} \xi^a \\ \chi^+_{\dot{a}} \end{pmatrix}$$

$$\bar{\Psi}^c \equiv C\bar{\Psi}^T = \begin{pmatrix} \xi_a \\ \chi^{+\dot{a}} \end{pmatrix}.$$

Hence we have $C^{-1}\Psi C = \Psi^c$

Numerically $C^T = C^\dagger = C^{-1} = -C$

$$C^{-1}\gamma^\mu C = \begin{pmatrix} \varepsilon^{ab} & \\ & \varepsilon_{\dot{a}\dot{b}} \end{pmatrix}\begin{pmatrix} 0 & \sigma^\mu_{b\dot{c}} \\ \bar{\sigma}^{\mu\dot{b}c} & 0 \end{pmatrix}\begin{pmatrix} \varepsilon_{ce} & \\ & \varepsilon^{\dot{c}\dot{e}} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \varepsilon^{ab}\sigma^\mu_{b\dot{c}}\varepsilon^{\dot{c}\dot{e}} \\ \varepsilon_{\dot{a}\dot{b}}\bar{\sigma}^{\dot{b}c}\varepsilon_{ce} & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -\bar{\sigma}^{\mu a\dot{e}} \\ -\sigma^\mu_{\dot{a}e} & 0 \end{pmatrix}$$

but $$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu_{e\dot{a}} \\ \bar{\sigma}^{\mu\dot{e}a} & 0 \end{pmatrix}$$

Hence $C^{-1}\gamma^\mu C = -(\gamma^\mu)^T$.

For Majorana field $\mathcal{L} = \frac{i}{2}\bar{\psi}\gamma^\mu\partial_\mu\psi - \frac{1}{2}m\bar{\psi}\psi$

but here we yet impose the majorana condition.

Skip due to time.

$$\psi^c = \psi.$$

we can use the above relations to write

$$\bar{\psi} = \psi^T C$$

$$\mathcal{L}_{majorana} = \frac{i}{2}\psi^T C\gamma^\mu\partial_\mu\psi - \frac{1}{2}m\psi^T C\psi.$$

$$\psi^c = C\bar{\psi}^T$$

$$C^T\bar{\psi}^c = C^{-1}C\bar{\psi}^T$$

$$\bar{\psi} = (C^T\bar{\psi}^c)^T = \psi^{cT}C$$

$$= \psi^T C$$

Here we can also find another important γ_5 matrix, which do not have Lorentz vector index

$$\gamma_5 \equiv \begin{pmatrix} -\delta_a{}^c & \\ & +\delta^{\dot{a}}{}_{\dot{c}} \end{pmatrix} \quad \text{such that}$$

Projection operator

$$P_L \equiv \frac{1}{2}(\mathbb{1}_{4\times4} - \gamma_5) = \begin{pmatrix} \delta_a{}^c & 0 \\ 0 & 0 \end{pmatrix}$$

$$P_R \equiv \frac{1}{2}(\mathbb{1} + \gamma_5) = \begin{pmatrix} 0 & 0 \\ 0 & \delta^{\dot{a}}{}_{\dot{c}} \end{pmatrix}.$$

For a Dirac field, we then have

$$P_L \Psi = \begin{pmatrix} \chi_a \\ 0 \end{pmatrix}, \quad P_R \Psi = \begin{pmatrix} 0 \\ \xi^{\dagger \dot{a}} \end{pmatrix}$$

γ_5 is an invariant symbol as (and we do not write it as γ^5 or γ_5)

$$\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = -\frac{i}{24}\varepsilon_{\mu\nu\rho\sigma}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma$$ [1]

Now, finally, let's come back to Lorentz transformations under our current conventions.

$$U(\Lambda)^{-1}\psi_a(x)\,U(\Lambda) = L(\Lambda)_a{}^c\,\psi_c(\Lambda^{-1}x)$$

$$U(\Lambda)^{-1}\psi^\dagger_{\dot{a}}(x)\,U(\Lambda) = R(\Lambda)_{\dot{a}}{}^{\dot{c}}\,\psi^\dagger_{\dot{c}}(\Lambda^{-1}x).$$

$$L = (1+\delta\omega)_a{}^c = \delta_a{}^c + \frac{i}{2}\delta\omega_{\mu\nu}(S_L^{\mu\nu})_a{}^c$$

$$R = (1+\delta\omega)_{\dot{a}}{}^{\dot{c}} = \delta_{\dot{a}}{}^{\dot{c}} + \frac{i}{2}\delta\omega_{\mu\nu}(S_R^{\mu\nu})_{\dot{a}}{}^{\dot{c}}$$

and

$$(S_L^{\mu\nu})_a{}^c = +\frac{i}{4}(\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu)_a{}^c$$

$$(S_R^{\mu\nu})_{\dot{a}}{}^{\dot{c}} = -\frac{i}{4}(\bar{\sigma}^\mu\sigma^\nu - \bar{\sigma}^\nu\sigma^\mu)_{\dot{a}}{}^{\dot{c}}$$

we can then define

$$\frac{i}{4}[\gamma^\mu,\gamma^\nu] = \begin{pmatrix} +(S_L^{\mu\nu})_a{}^c & \\ & -(S_R^{\mu\nu})^{\dot{a}}{}_{\dot{c}} \end{pmatrix} \equiv S^{\mu\nu}$$

1 Sign convention: this line and p. 12's $\mathrm{Tr}[\gamma_5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = -4i\epsilon^{\mu\nu\rho\sigma}$ both need Srednicki's $\epsilon^{0123} = +1$, $\epsilon_{0123} = -1$ (§34); the other sign flips both. Check: $\gamma^0\gamma^1\gamma^2\gamma^3 = -i\gamma_5$. "Invariant symbol": $[\gamma_5, S^{\mu\nu}] = 0$, so $P_L\Psi$ and $P_R\Psi$ transform separately, as $(2,1)$ and $(1,2)$.

1 Why $\bar\Psi = \Psi^\dagger\beta$, not $\Psi^\dagger$: $D(\Lambda) = \mathrm{diag}(L, (L^\dagger)^{-1})$ is not unitary (boosts), so $\Psi^\dagger\Psi$ is not a scalar. But $\beta D^\dagger \beta = D^{-1}$, so $\bar\Psi \to \bar\Psi D^{-1}$, and $D^{-1}\gamma^\mu D = \Lambda^\mu{}_\nu \gamma^\nu$: $\bar\Psi\Psi$ is a scalar, $\bar\Psi\gamma^\mu\partial_\mu\Psi$ too, so $\mathcal{L}$ (p. 6) is invariant. γ^0 equals β (p. 5) only as numbers.

Then we have

$$U(\Lambda)^{-1}\Psi_R(x)U(\Lambda) = D(\Lambda)\Psi_R(\Lambda^{-1}x)$$

and $D(1+\delta\omega) = 1 + \frac{i}{2}\delta\omega_{\mu\nu}S^{\mu\nu}$ (1)

and here the "$-$" for S_R is taken care by the $\dot{i}$ to $\dot{i}^{\dot{c}}$.

$S^{\mu\nu}$ is the generator for L.I. for Dirac field $\Psi_R(x)$.

Gamma Matrices Gymnastics.

$$\{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}$$

$$\gamma_5^2 = 1$$

$$\boxed{\{\gamma^\mu, \gamma_5\} = 0}.$$

$$\mathrm{Tr}\,1 = 4.$$

Now $\mathrm{Tr}[\text{odd no. of } \gamma_5] = 0$

as $\mathrm{Tr}[\gamma^{\mu_1}\cdots\gamma^{\mu_n}] = \mathrm{Tr}[\gamma_5^2\gamma^{\mu_1}\cdots\gamma_5^2\gamma^{\mu_n}]$

$= \mathrm{Tr}[\not{\ }(\gamma_5\gamma^{\mu_1}\gamma_5)\cdots(\gamma_5\gamma^{\mu_n}\gamma_5)]$

$= (-1)^n\,\mathrm{Tr}[\gamma^{\mu_1}\cdots\gamma^{\mu_n}]$.

$\gamma^\mu \partial_\mu = \not{\partial}$
$\gamma^\mu p_\mu = \not{p}$

Similarly

$$\mathrm{Tr}[\gamma^5 \ldots \underbrace{\gamma^{\mu_1} \ldots \gamma^{\mu_n}}_{\text{odd \#}}] = 0.$$

Also

$$\mathrm{Tr}[\gamma^\mu \gamma^\nu] = \frac{1}{2}\mathrm{Tr}[\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu]$$
$$= -4g^{\mu\nu}.$$

$$\mathrm{Tr}[\not{a}\not{b}] = -4(a\cdot b).$$

$$\mathrm{Tr}[\not{a}\not{b}\not{c}\not{d}] = 4[(ad)(bc) - (ac)(bd) + (ab)(cd)]$$

↑

this is ~~pure~~ proven by

$$\not{a}\not{b} = -\not{b}\not{a} - 2(a\cdot b)$$

$$\mathrm{Tr}[\not{a}\not{b}\not{c}\not{d}] = -\mathrm{Tr}[\not{b}\not{a}\not{c}\not{d}] - 2(ab)\,\mathrm{Tr}[\not{c}\not{d}]$$

$\mathrm{Tr}[\not{c}\not{d}] = 4(cd)$

$$-\mathrm{Tr}[\not{b}\not{c}\not{a}\not{d}] \quad (1)$$
$$-2(ac)\,\mathrm{Tr}[\not{b}\not{d}]$$

↓

. . .

↓

. . .

$$= -\mathrm{Tr}[\not{a}\not{b}\not{c}\not{d}] + \ldots$$

1 The chain in full: moving $\not{a}$ right, $\mathrm{Tr}[\not{a}\not{b}\not{c}\not{d}] = -2(ab)\mathrm{Tr}[\not{c}\not{d}] + 2(ac)\mathrm{Tr}[\not{b}\not{d}] - 2(ad)\mathrm{Tr}[\not{b}\not{c}] - \mathrm{Tr}[\not{b}\not{c}\not{d}\not{a}]$; by cyclicity the last term is $-\mathrm{Tr}[\not{a}\not{b}\not{c}\not{d}]$, which gives the boxed result. The arrow expands $\mathrm{Tr}[\not{b}\not{a}\not{c}\not{d}]$, without the minus it carries.

One can then perform arbitrary even traces.

Now, γ_5. $\{\gamma_5, \gamma^\mu\} = 0$ and $\gamma_5^2 = 1$

Then and trace with γ_5s is only one γ_5.

Also $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $Tr[\gamma_5] = 0$

$$Tr[\gamma_5 \gamma^\mu \gamma^\nu] = 0$$

~~$Tr[\gamma_5 \gamma^\mu \gamma^\nu] = -Tr[\gamma_5 \gamma^\nu \gamma^\mu] + 2Tr[\gamma_5] g^{\mu\nu}$~~

$\mu = \nu$, $\gamma^2 = 1$, $Tr[\gamma_5] = 0$

$\mu \neq \nu$ $\begin{cases} 0,i & Tr[(-1,)(1,1)(\vec{-}\vec{2})] = 0 \\ i,j & Tr[(-1,)(\vec{-}\vec{2})(\vec{-}\vec{2})] = Tr[\vec{2}\cdot\vec{2}] = 0 \end{cases}$

And $Tr[\gamma_5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = -4i\varepsilon^{\mu\nu\rho\sigma}$

Another set of useful trick/identity is

$$\gamma^\mu \gamma_\mu = g_{\mu\nu} \gamma^\mu \gamma^\nu = \frac{1}{2} g_{\mu\nu} \{\gamma^\mu, \gamma^\nu\} = -d.$$

need for dim-reg.

$$\gamma^\mu \not{a} \gamma_\mu = \gamma^\mu (-\gamma_\mu \not{a} - 2a_\mu)$$
$$= (d-2)\not{a}$$

$$\gamma^\mu \not{a} \not{b} \gamma_\mu = 4(a\cdot b) - (d-4)\not{a}\not{b}$$

- - - -

Let's do path integral, Recall for real scalar

$$\mathcal{L}_0 = -\frac{1}{2}\partial^\mu\varphi\,\partial_\mu\varphi - \frac{1}{2}m\varphi^2$$

$$= -\frac{1}{2}\varphi(-\partial^2+m^2)\varphi - \frac{1}{2}\partial_\mu(\varphi\partial^\mu\varphi)$$

We had

$$\langle 0|T\varphi(x_1)\ldots|0\rangle = \frac{1}{i}\frac{\delta}{\delta J(x_1)}\cdots Z_0(J)\Big|_{J=0}$$

$$Z_0(J) = \int \mathcal{D}\varphi\, \exp\left[i\int d^4x\,(\mathcal{L}+J\varphi)\right]$$

we used the $i\varepsilon$ to make sure the initial and final states are vacuum state (as others it exponentially suppresses and state with non-zero energy)

And then (perform the Gaussian integration...)

$$Z_0(J) = \mathcal{N}\exp\left[\frac{i}{2}\int d^4x\,d^4y\,J(x)\Delta(x-y)J(y)\right]$$

$$\Delta(x-y) = \int \frac{d^4k}{(2\pi)^4}\frac{e^{ik(x-y)}}{k^2+m^2-i\varepsilon}$$

with is the inverse of the Klein-Gordon wave operator

$$(-\partial_x^2+m^2)\Delta(x-y) = \delta^4(x-y).$$

And for complex scalar field (shown in your HWs)

$$\mathcal{L}_0 = -\partial^\mu \varphi^\dagger \partial_\mu \varphi - m^2 \varphi^\dagger \varphi.$$

$$= -\varphi^\dagger(-\partial^2 + m^2)\varphi - \partial_\mu(\varphi^\dagger \partial^\mu \varphi).$$

such that the path integral

$$Z_0[J^\dagger, J] = \int \mathcal{D}\varphi^\dagger \mathcal{D}\varphi \exp\left[i\int d^4x (\mathcal{L}_0 + J^\dagger\varphi + J\varphi^\dagger)\right]$$

$$= \exp\left[i\int d^4x\, d^4y\, J^\dagger(x)\Delta(x-y)J(y)\right].$$

Cool

$$\langle 0|T[\varphi(x_1)\cdots\varphi^\dagger(y_1)\cdots|0\rangle = \frac{1}{i}\frac{\delta}{\delta J^\dagger(x_1)}\cdots\frac{\delta}{i\delta J(y_1)}\cdots Z_0[J^\dagger, J]\Big|_{J=J^\dagger=0}$$

So now for Dirac Fermions, we do things similarly, but now pay special attention that $\psi(x)$ and $\bar{\psi}(x)$ are spinors ~~and~~ (not scalars that and be moved around), and the source terms are also spinors $\bar{\eta}(x)$ and $\eta(x)$ such that $\mathcal{L}$ remains a scalar quantity, [1]

1 Why $\eta, \bar{\eta}$ anticommute: $\bar{\eta}\Psi$ sits in the exponent beside $\mathcal{L}$, so it must be an ordinary commuting number, and Ψ anticommutes. Then $\frac{\delta}{\delta\bar{\eta}(x)}\frac{\delta}{\delta\eta(y)} = -\frac{\delta}{\delta\eta(y)}\frac{\delta}{\delta\bar{\eta}(x)}$: fermionic $\langle\mathcal{T}\ldots\rangle$ flip sign when two fields are swapped. That is where the relative minus signs between diagrams come from (LecNote18 p. 2).

also, there are anti-commutation relations. So

$$\frac{\delta}{\delta\eta(x)}\int d^4y\,[\bar\eta(y)\Psi(y)+\bar\Psi(y)\eta(y)] = -\bar\Psi(x)$$

$\bar\eta$: row vector, Ψ: column vector. ↑ from $\eta(x)$ passing through $\bar\Psi(y)$ anti-commutation.

$$\frac{\delta}{\delta\bar\eta(x)}\int d^4y\,[\bar\eta(y)\Psi(y)+\bar\Psi(y)\eta(y)] = -\Psi(x)$$

Now the free Dirac field

$$\mathcal{L}_0 = i\bar\Psi\partial\!\!\!/\Psi - m\bar\Psi\Psi$$

$$= \cancel{-\bar\Psi(\partial\!\!\!/ + m)i} - \bar\Psi(-i\partial\!\!\!/ + m)\Psi$$

We shall have

$$\langle 0|T\,\Psi_{\alpha_1}(x_1)\cdots\bar\Psi_{\beta_1}(y_1)\ldots|0\rangle$$

$$= \frac{1}{i}\frac{\delta}{\delta\bar\eta_{\alpha_1}(x_1)}\cdots\frac{i}{\mp}\frac{\delta}{\delta\eta_{\beta_1}(y_1)}\cdots\cancel{\mp}\,Z_0(\bar\eta,\eta)\Big|_{\bar\eta=\eta=0}$$

$(-1)^{\#}/i$

$$Z_0(\bar\eta,\eta) = \int D\Psi D\bar\Psi\,\exp\left[i\int d^4x\,(\mathcal{L}_0 + \bar\eta\Psi + \bar\Psi\eta)\right]$$

$$= \exp\left[-i\int d^4x\,d^4y\,\bar\eta(x)\,S(x-y)\,\eta(y)\right]$$ (1)

1 Why this Z_0: with $D = -i\partial\!\!\!/ + m$, shift $\Psi \to \Psi + S\eta$, $\bar\Psi \to \bar\Psi + \bar\eta S$ ($DS = \delta^4$): the exponent becomes $i\int(-\bar\Psi D\Psi + \bar\eta S\eta)$. The leftover integral is a constant (a determinant, p. 20), removed by $Z_0(0,0) = 1$. No $\frac{1}{2}$: $\Psi, \bar\Psi$ are independent, like $\varphi, \varphi^\dagger$ (p. 14). $\tilde S = (\not p + m)^{-1}$; p. 21 does this for Grassmann numbers.

Here $S(x-y)$ is the Feynman propagator.

$$S(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{(-\not{p}+m)}{p^2+m^2-i\epsilon} e^{ip(x-y)}$$

which is the inverse of the Dirac wave operator:

$$(-i\not{\partial}_x + m) S(x-y) = \delta^4(x-y).$$

Skip the following due to time

The above can be done more rigorously, see Sec. 44. Here I sketch the key components.

Let me introduce Grassmann numbers. ψ_i

(anti commuting numbers)

$$\{\psi_i, \psi_j\} = 0, \quad i,j = 1, \ldots, n.$$

For $n=1$, $\psi_1^2 + \psi_1^2 = 0 \rightarrow \psi_1^2 = 0$

So any function $f(\psi) = a + \psi b$ as high orders vanish.

Let's demand $f(\psi)$ is a regular commuting number/function then we need to have $\{\psi, b\} = \{b, b\} = 0$

and hence

$$f(\psi) = a + \psi b = a - b\psi.$$

then we can define derivatives

$$\partial_\psi f(\psi) = b$$

$$f(\psi) \overleftarrow{\partial}_\psi = -b$$

Similarly, integrals demanding

$$\int_{-\infty}^{+\infty} dx\, c f(x) = c \int_{-\infty}^{\infty} dx\, f(x)$$

$$\int_{-\infty}^{+\infty} dx\, f(x+a) = \int_{-\infty}^{\infty} dx\, f(x)$$

$\int dx\, c f(x+d) = \int dx\, c f(x)$

$\Rightarrow \int dx\, ca + cdb = \int dx\, ca \Rightarrow \int dx\, cdb = 0$

tells us that

consistent definition is

$$\int_{-\infty}^{+\infty} dx\, f(x) = b \,, \quad \int_{-\infty}^{\infty} dx = 0$$

Similarly for $i \to n$ for ψ_i

$$\frac{\partial}{\partial \psi_j} f(\psi) = b_j + \psi_i c_{ji} + \cdots + \frac{1}{(n-1)!} \psi_{i_2} \cdots \psi_{i_n}$$

$$\underline{d_{j i_2 \cdots i_n}}$$

↑ totally antisymmetric symbol

$$d_{i_1 \cdots i_n} = d\, \varepsilon_{i_1 \cdots i_n}$$

↑ Levi-Civita

$$\Rightarrow \int d^n\psi\, f(\psi) = d.$$

$$\int d\psi_n\, d\psi_{n-1} \cdots d\psi_1$$

Also, consistently

$$\int d\psi_j = 0, \quad \int d\psi_i\, \psi_i = \delta_{ij}$$

Now, if we change coordinates linearly

$\psi_i = J_{ij}\psi_j$, J is a normal matrix

$$f(\psi) = a + \cdots + \frac{1}{n!}(J_{i_1 j_1}\psi'_{j_1})\cdots(J_{i_n j_n}\psi'_{j_n})\varepsilon_{i_1\cdots i_n} d.$$

$$\prod_j d\psi'_j \det J \; \varepsilon_{i_1\cdots i_n}$$

$$f(\psi) = a + \cdots + \frac{1}{n!}\psi'_{j_1}\cdots\psi'_{j_n}(\det J)\varepsilon_{j_1\cdots j_n} d.$$

$$\rightarrow \int d^n\psi\, f(\psi) = (\det J)^{-1}\int d^n\psi'\, f(\psi').$$

Recall for normal numbers, we have (commuting)

$$\int d^n x\, f(x) = (\det J)^{+1}\int d^n x'\, f(x').$$

Grassmannian numbers are different.

Now, back to fermions, we can define the following

$$\chi \equiv \frac{1}{\sqrt{2}}(\psi_1 + i\psi_2)$$

$$\bar{\chi} = \frac{1}{\sqrt{2}}(\psi_1 - i\psi_2)$$

$$\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ i & -i\end{pmatrix}\begin{pmatrix}\bar{x}\\ x\end{pmatrix}.$$

so $d^2\psi = d\psi_2\, d\psi_1 = (-i)^{-1}\, dx\, d\bar{x}$

also $\psi_1\psi_2 = -i\bar{x}x$, hence

$$\int dx\, d\bar{x}\ \bar{x}x = (-i)(-i)^{-1}\int d\psi_2\, d\psi_1\ \psi_1\psi_2 = 1.$$

thus, for a general function

$$f(x,\bar{x}) = a + xb + \bar{x}c + \bar{x}xd.$$

$$\int dx\, d\bar{x}\ f(x,\bar{x}) = d$$

$$\int dx\, d\bar{x}\ \exp(\bar{x}mx) = m$$

(only by linear order matters)

Similarly, for

~~$d^n x\, d^m \bar{x}$~~ $d^n x\, d^n \bar{x} = (\det J)^{-1}(\det K)^{-1} d^n x'\, d^n \bar{x}'$

for $x_i \to J_{ij} x'_j$

$\bar{x}_i \to K_{ij}\bar{x}'_j$

Now

if we evaluate a general matrix M.

$$\int d^n x\, d^n \bar{x}\ \exp(\bar{x} M x)$$

↓ not diagonal

but M can be diagonalized via

$$M = V M U = M_{diag}.$$

$$V V^\dagger = V^\dagger V = 1$$
$$U^\dagger U = U U^\dagger = 1$$

$$\int d^n x\, d^n \bar{x}\ \exp(\bar{x} M x)$$

$$= \int d^n x\, d^n \bar{x}\ \exp(\bar{x} V^\dagger V M U U^\dagger x)$$

$$= \int d^n x'\, d^n \bar{x}'\ \exp(\bar{x}' M_{diag} x')\ \det(V^\dagger)\det(U^\dagger)$$

$$= \det(V)^{-1} \det(U)^{-1}\ \cancel{M_{diag}}\ \det M_{diag}$$

$$\det M = \det(V^{-1} M_{diag} U^{-1})$$
$$= \det V^{-1}\ \det M_{diag}\ \det U^{-1}$$

$$\rightarrow \int d^n x\, d^n \bar{x}\ \exp(\bar{x} M x) = \det M.$$

and again, for commuting numbers

$$\int d^n z\, d^n \bar{z}\ \exp(-z^\dagger M z) = (2\pi)^n (\det M)^{-1}$$

can you prove it?

So now

$$\int d^n x\, d^n \bar{x}\, \exp\left(\bar{x} M x + \bar{\eta} x + \bar{x} \eta\right)$$

$$= (\det M) \exp\left(-\bar{\eta} M^{-1} \eta\right)$$

as the numerator is a complex sepreae via

$$x \to x - M^{-1}\eta, \quad \bar{x} \to \bar{x} - \bar{\eta} M^{-1}$$

this justifies our path-integral.

Skip above

So now let's talk about interacting theory & Feynmen Rules.

adding $\mathcal{L}_1 = g \varphi \bar{\Psi} \Psi$.

↑ real scalar.

tree-level only.

We see at $d=4$, g is dimensionless.

This interacting still preserves charge. as

$\Psi \to e^{-i\alpha} \Psi$ makes $\mathcal{L}_1$ invariant.

1 Should read: this is Yukawa theory (§45); scalar QED is a charged scalar coupled to A^μ. Order per p. 15: $\exp[ig\int \frac{1}{i}\frac{\delta}{\delta J}\, i\frac{\delta}{\delta\eta_\alpha}\frac{1}{i}\frac{\delta}{\delta\bar\eta_\alpha}]$, as $\bar\Psi_\alpha\Psi_\alpha$; the order written gives $-g$.

This theory is called "Scalar QED" (1)

Now $Z(\bar\eta,\eta,J) \propto \exp\left[ig\int d^4x \left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)\left(\frac{1}{i}\frac{\delta}{\delta\bar\eta_\alpha(x)}\right)\left(\frac{(-1)}{i}\frac{\delta}{\delta\eta(x)}\right)\right]$

$\uparrow\ \updownarrow\ \uparrow$

$\Psi\ \bar\Psi\ \varphi.$

2 Should read $\exp[\frac{i}{2}\int J\Delta J]$: φ is real (p. 13); without the $\frac{1}{2}$ every scalar line doubles. The scalar mass is independent of the fermion's: write M in Δ (LecNote20 writes m_φ).

$Z_0(\bar\eta,\eta,J)$.

where

$$Z_0(\bar\eta,\eta,J) = \exp\left[i\int d^4x\, d^4y\ \bar\eta(x) S(x-y)\eta(y)\right]$$

$$\times \exp\left[i\int d^4x\, d^4y\ J(x)\Delta(x-y)J(y)\right] (2)$$

with $$S(x-y) = \int \frac{d^4p}{(2\pi)^4}\frac{(-\not p + m)}{p^2+m^2-i\epsilon} e^{-ip(x-y)}$$

$$\Delta(x-y) = \int \frac{d^4p}{(2\pi)^4}\frac{1}{p^2+m^2-i\epsilon} e^{-ip(x-y)}.$$

Similarly to the scalar field theory case, we have

$$Z(\bar\eta,\eta,J) = \exp\left[iW(\bar\eta,\eta,J)\right]$$

$\uparrow$

connected Feynman diagrams (with sources).